



2001

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet, formulae sheets and Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 13, 15, 17 and 21

Total marks – 100

Section I Pages 2–23

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–26
- Allow about 1 hour and 45 minutes for this part

Section II Pages 25–31

25 marks

- Attempt ONE question from Questions 27–31
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

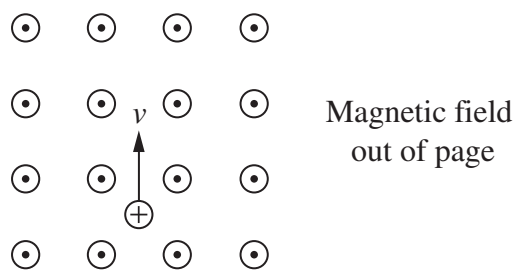
If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

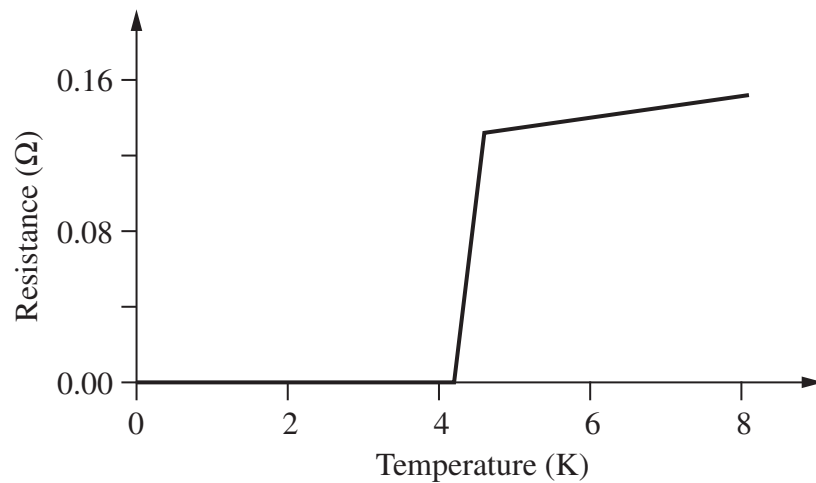
A ☒ B ☒ C ☐ D ☐
correct
↖

- 1 A person has a mass of 70.0 kg. What is the weight of the person at the Earth's surface?
- (A) 70.0 kg
(B) 70.0 N
(C) 686 kg
(D) 686 N
- 2 At a particular moment, a positively charged particle is moving with velocity v in a magnetic field as shown.



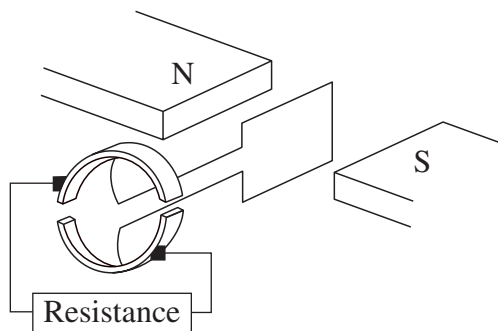
- At this moment, what is the direction of the force on the positively charged particle?
- (A) To the right
(B) To the left
(C) Into the page
(D) Out of the page

- 3 The resistance of mercury at various temperatures is shown in the graph.

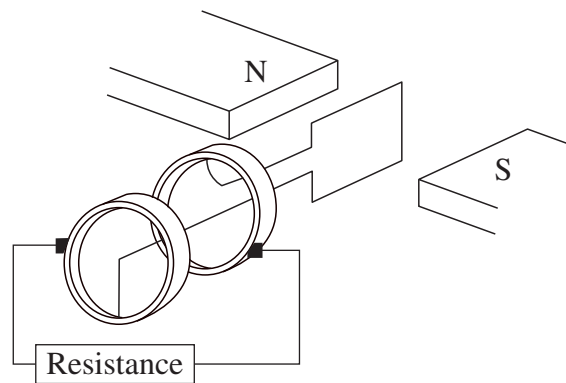


Between which two temperatures does mercury always act as a superconductor?

- (A) 0 K and 4.2 K
 - (B) 4.2 K and 4.5 K
 - (C) 4.5 K and 8.0 K
 - (D) 0 K and 8.0 K
- 4 Two types of generator are shown.



Generator 1

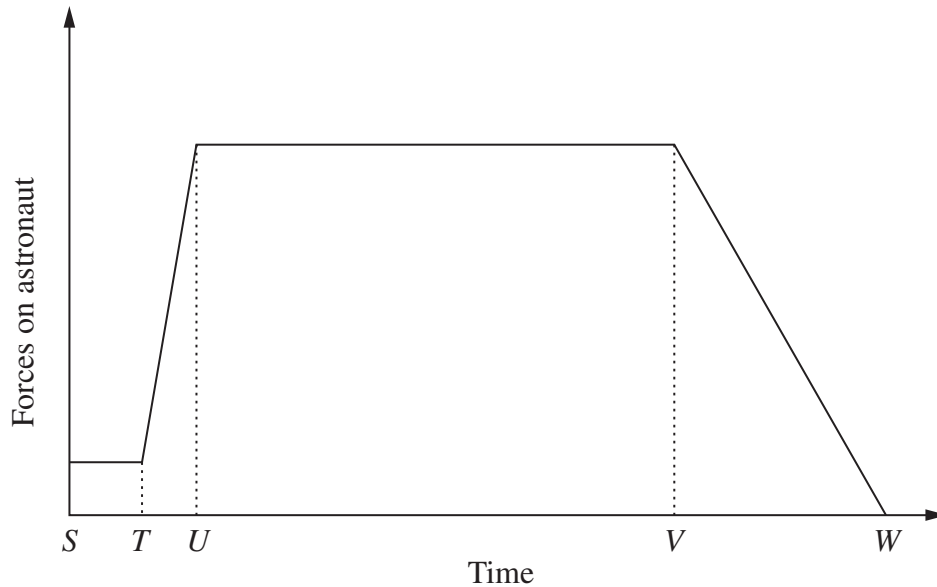


Generator 2

What type of current is produced by each generator when connected to an external resistance?

- (A) Both produce d.c.
- (B) Both produce a.c.
- (C) Generator 1 produces d.c. and Generator 2 produces a.c.
- (D) Generator 1 produces a.c. and Generator 2 produces d.c.

- 5 The graph shows the forces experienced by an astronaut during a rocket launch into a stable orbit.



In which time interval was the acceleration of the rocket the greatest?

- (A) $S-T$
 - (B) $T-U$
 - (C) $U-V$
 - (D) $V-W$
- 6 The signal from a microwave transmitter can be thought of as a beam of photons. The photons from a particular transmitter have a wavelength of 3.5×10^{-2} m.

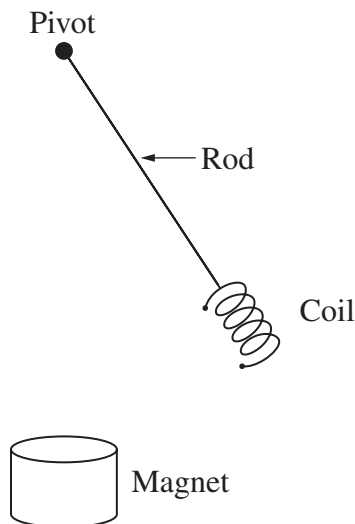
What is the approximate energy of each photon?

- (A) 7.73×10^{-44} J
- (B) 5.68×10^{-24} J
- (C) 2.32×10^{-35} J
- (D) 1.89×10^{-32} J

- 7 An astronaut is standing on Mars. The astronaut throws an object of mass 0.30 kg vertically upward at an initial speed of 9.0 m s^{-1} . It reaches a maximum height of 11 metres .

What is the magnitude of the acceleration of the object?

- (A) 1.4 m s^{-2}
(B) 3.7 m s^{-2}
(C) 9.0 m s^{-2}
(D) 9.8 m s^{-2}
- 8 A light rod has a coil of insulated copper wire fixed at one end and is pivoted at the other end. The result is a pendulum which is free to swing back and forth. A magnet is placed underneath this pendulum. The arrangement is shown in the diagram.



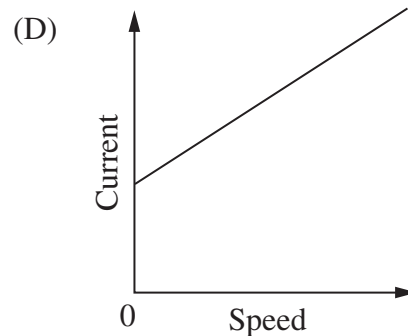
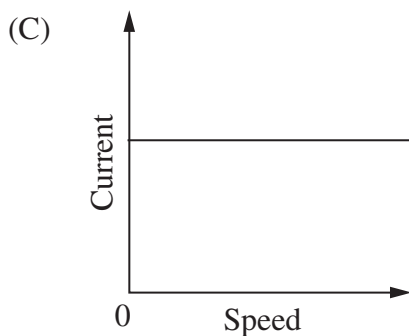
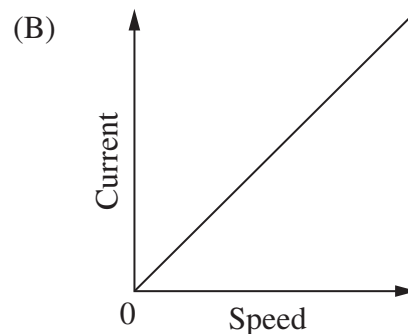
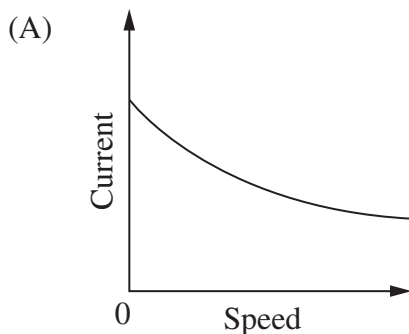
The pendulum is pulled back and then allowed to swing. Which of the following would cause the pendulum to come to rest most quickly?

- (A) Replacing the magnet with a stronger one
(B) Shortening the pendulum
(C) Replacing the rod with a heavier one
(D) Connecting the ends of the coil by a piece of copper wire

- 9 Which is the most suitable means of reliable and continuous communication between an orbiting satellite and Earth?
- (A) Light from a green laser
 - (B) Microwaves
 - (C) Radio waves
 - (D) Sound waves

- 10 An electric motor is connected to a power supply of constant voltage. The motor is allowed to run at different speeds by adjusting a brake.

Which graph best shows how the current through the motor varies with speed?



- 11 A transformer has a primary coil with 60 turns and a secondary coil with 2300 turns.

If the primary voltage to the transformer is 110 V, what is the secondary voltage?

- (A) 2.4×10^{-4} V
- (B) 2.4×10^2 V
- (C) 1.3×10^3 V
- (D) 4.2×10^3 V

- 12 Which of the following statements best describes the reason why some materials become superconducting at very low temperatures?
- (A) The ions in the superconductor form a regular crystal lattice. There are long channels through the lattice along which the electrons can pass without colliding with the lattice.
 - (B) Vibrations of the crystal lattice are so small that they do not interfere with the motion of the electrons.
 - (C) Electrons in a superconductor have very low energy. Their energy is so low that they cannot transfer energy to the crystal lattice in a collision.
 - (D) Electrons ‘pair up’. These electron pairs pass through the crystal lattice of the superconductor without losing energy in collisions with the lattice.
- 13 A rocket car moves on a straight horizontal track. Half of the initial mass of the rocket car is propellant. During the run, propellant is consumed at a constant rate and ejected at a constant nozzle velocity.

Which of the following best describes the force propelling the rocket car, and the magnitude of the acceleration of the rocket car while the propellant is being ejected?

	<i>Force</i>	<i>Acceleration</i>
(A)	constant	constant
(B)	increasing	constant
(C)	constant	increasing
(D)	increasing	increasing

- 14 Two straight metal rods, P and Q , have the same length. They are each pivoted at one end and rotated with the same angular velocity so that they sweep out horizontal circular paths as shown in diagrams X and Y . A constant current I is flowing along each rod, as shown.

In diagram X , a constant magnetic field is applied *at right angles to the plane* of the circular path. In diagram Y , a uniform magnetic field of the same magnitude is applied *in the plane* of the circular path.

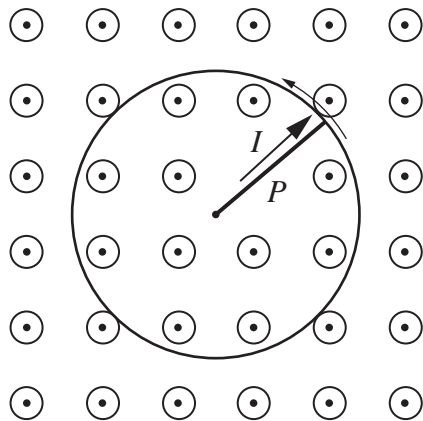


Diagram X

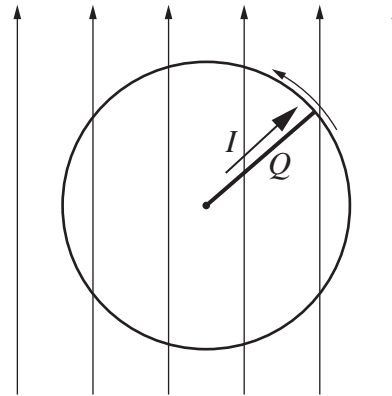


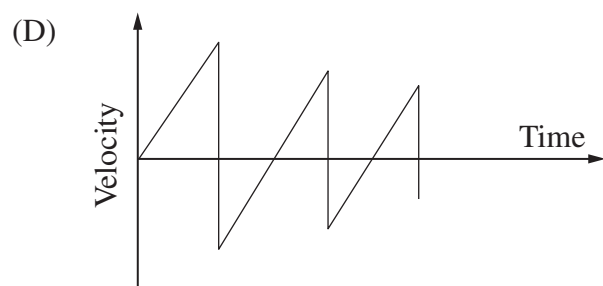
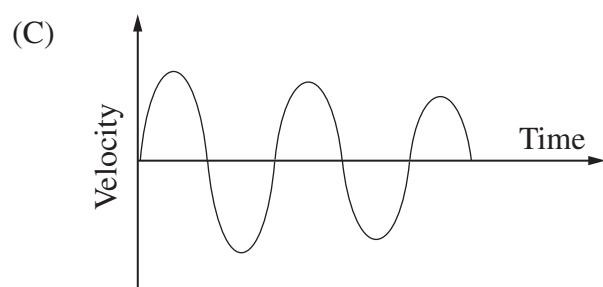
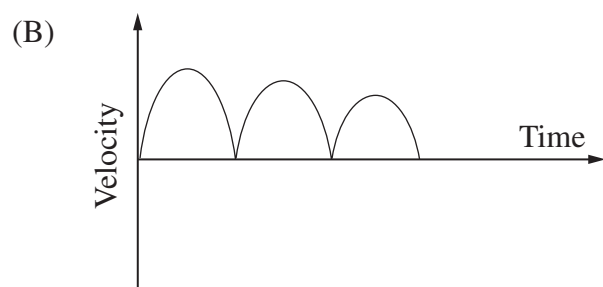
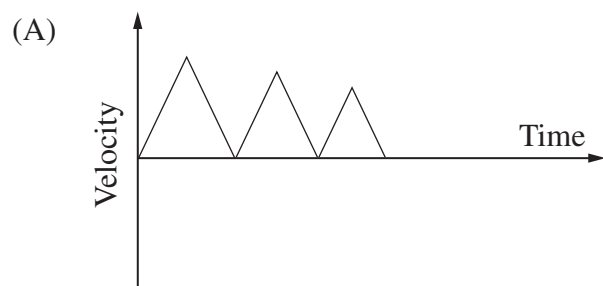
Diagram Y

Which of the following statements about the forces acting on rod P and rod Q is correct?

- (A) The magnitude of the force on P is exactly the same as the magnitude of the force on Q at all times.
- (B) The magnitude of the force on P is constant and the magnitude of the force on Q is zero.
- (C) The magnitude of the force on P is constant and the magnitude of the force on Q varies with time.
- (D) The magnitude of the force on P varies with time and the magnitude of the force on Q is constant.

- 15 A student releases a ball from eye level. The ball bounces several times.

Which velocity vs time graph best represents the ball's motion?



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Physics

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Centre Number

Section I (continued)

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Part B – 60 marks

Student Number

Attempt Questions 16–26

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Marks

Question 16 (4 marks)

Muons are very short-lived particles that are created when energetic protons collide with each other. A beam of muons can be produced by very-high-energy particle accelerators.

The high-speed muons produced for an experiment by the Fermilab accelerator are measured to have a lifetime of 5.0 microseconds. When these muons are brought to rest, their lifetime is measured to be 2.2 microseconds.

- (a) Name the effect demonstrated by these observations of the lifetimes of the muons. 1

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- (b) Calculate the velocity of the muons as they leave the accelerator. 3

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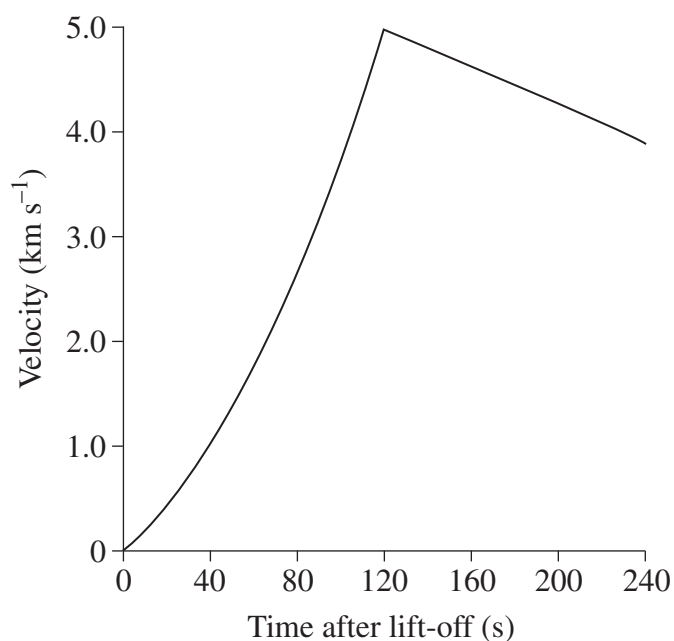
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Question 17 (6 marks)

A rocket was launched vertically to probe the upper atmosphere. The vertical velocity of the rocket as a function of time is shown in the graph.



- (a) Using either words or calculations, compare the acceleration of the rocket at $t = 20$ s with its acceleration at $t = 100$ s. **2**

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- (b) Account for the shape of the graph over the range of time shown. **4**

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Section I – Part B (continued)

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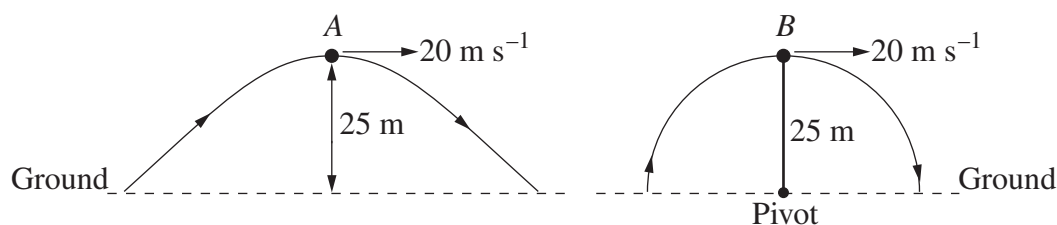
Student Number

Marks

Question 18 (6 marks)

A 30 kg object, *A*, was fired from a cannon in projectile motion. When the projectile was at its maximum height of 25 m, its speed was 20 m s^{-1} .

An identical object, *B*, was attached to a mechanical arm and moved at a constant speed of 20 m s^{-1} in a vertical half-circle. The length of the arm was 25 m.



Ignore air resistance.

- (a) Calculate the force acting on object *A* at its maximum height. 1

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- (b) Calculate the time it would take object *A* to reach the ground from its position of maximum height. 2

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- (c) Describe and compare the vertical forces acting on objects *A* and *B* at their maximum heights. 3

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Question 19 (4 marks)

How does Einstein's Theory of Special Relativity explain the result of the Michelson–Morley experiment? **4**

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Question 20 (4 marks)

The electrical supply network uses a.c. and a variety of transformers between the generating stations and the final consumer. **4**

Explain why transformers are used at various points in the network.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 21 (3 marks)

A fan that ventilates an underground mine is run by a very large d.c. electric motor. This motor is connected in series with a variable resistor to protect the windings in the coil.

3

When the motor is starting up, the variable resistor is adjusted to have a large resistance. The resistance is then lowered slowly as the motor increases to its operating speed.

Explain why no resistance is required when the motor is running at high speed, but a substantial resistance is needed when the motor is starting up.

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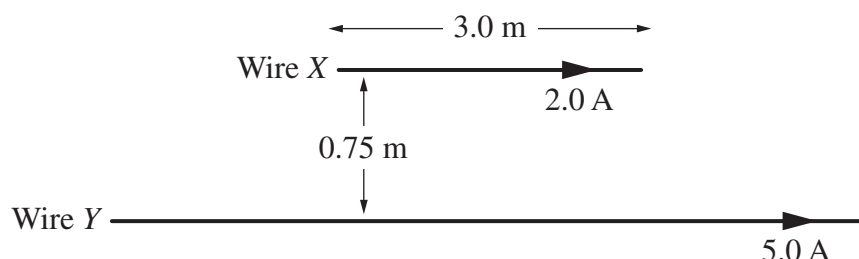
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Question 22 (7 marks)

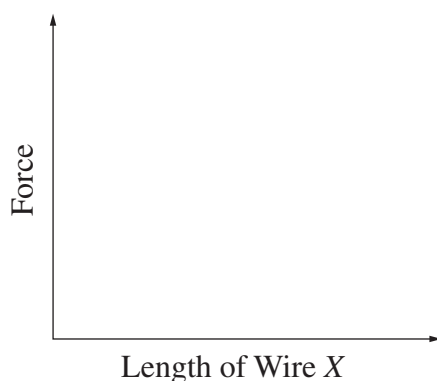
Two parallel wires are separated by a distance of 0.75 m. Wire X is 3.0 m long and carries a current of 2.0 A. Wire Y can be considered to be infinitely long and carries a current of 5.0 A. Both currents flow in the same direction along the wires.



- (a) What is the direction of the force that exists between the two wires? 1

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- (b) On the axes, sketch a graph that shows how the force between the two wires would vary if the length of Wire X was increased. 2



- (c) In your Physics course you have performed a first-hand investigation to demonstrate the motor effect. Explain how your results demonstrated that effect. 4

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Question 23 (6 marks)

Discuss the effects of the development of electrical generators on society and the environment.

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Centre Number

Section I – Part B (continued)

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Student Number

Marks

Question 24 (6 marks)

Sir William Bragg and his son Sir Lawrence Bragg shared the Nobel prize for physics in 1915 for their work on X-ray diffraction and crystal structure analysis.

- (a) Describe ONE way in which an understanding of crystal structure has impacted on science. 2

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- (b) Outline the methods of X-ray diffraction used by the Braggs to determine the structure of crystals. 4

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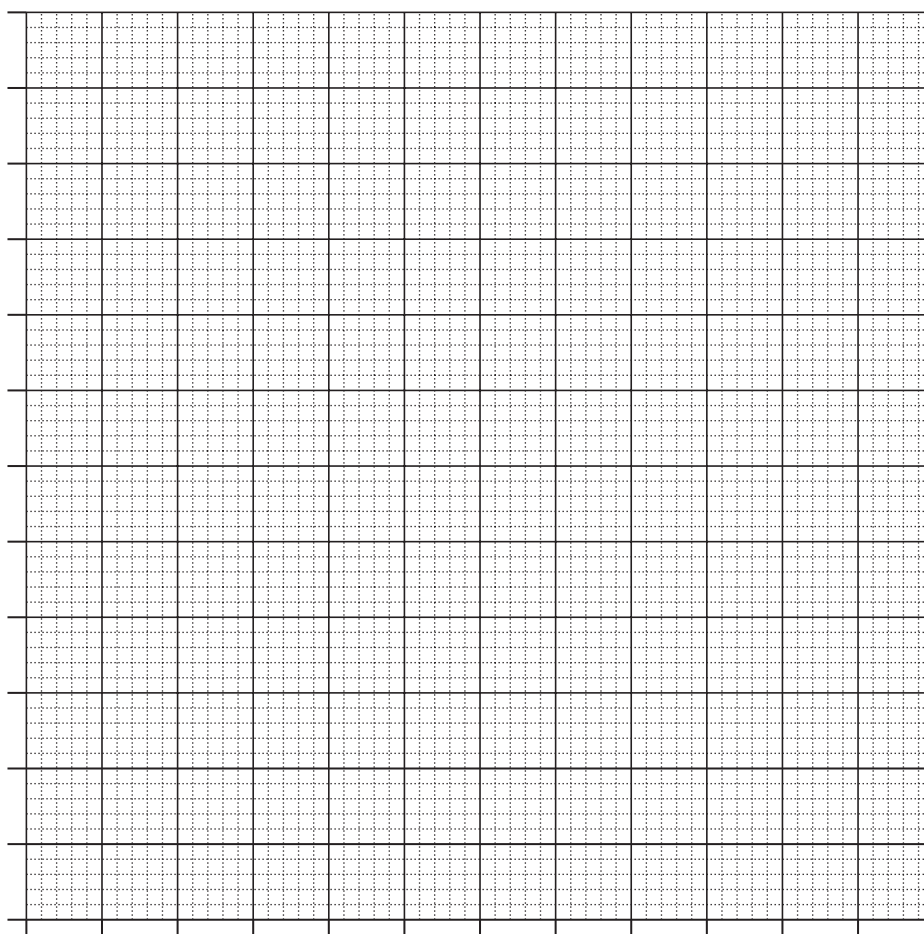
Question 25 (6 marks)

A student carried out an experiment on the photoelectric effect. The frequency of the incident radiation and the energy of the photoelectrons were both determined from measurements taken during the experiment.

The results obtained are shown in the table:

<i>Frequency of incident radiation</i> ($\times 10^{14}$ Hz)	<i>Energy of photoelectrons</i> ($\times 10^{-19}$ J)
6.9	1.22
8.2	1.70
9.1	3.70
9.9	3.05
10.6	3.38
11.8	3.91

- (a) Graph these results on the grid, including the line of best fit.

4

Question 25 continues on page 23

Question 25 (continued)

(b) How could the reliability of the experiment be improved?

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Question 26 (8 marks)

In the context of semiconductors, explain the concept of *electrons* and *holes*.

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Physics

Section II

25 marks

Attempt ONE question from Questions 27–31

Allow about 45 minutes for this section

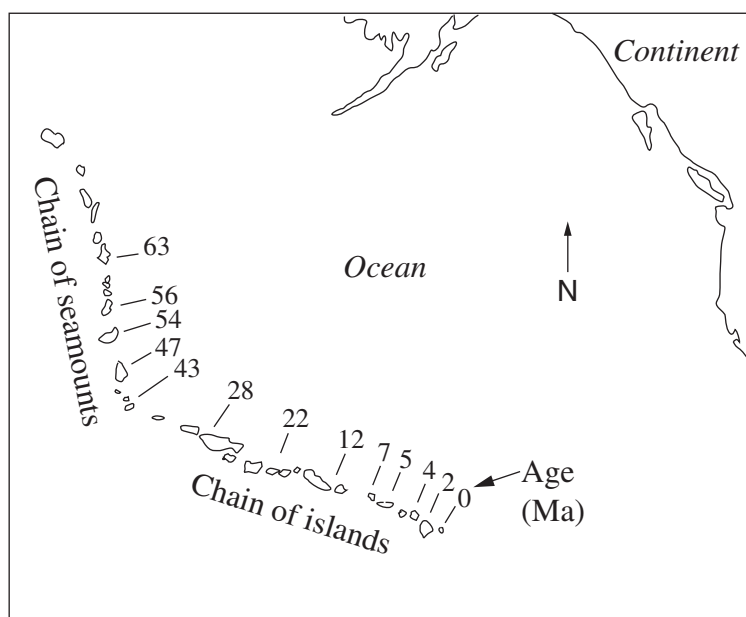
Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

	Pages
Question 27 Geophysics	26
Question 28 Medical Physics	27
Question 29 Astrophysics	28–29
Question 30 From Quanta to Quarks	30
Question 31 The Age of Silicon	31

Question 27 — Geophysics (25 marks)

- (a) (i) Name the instrument used in local gravity surveys. **1**
- (ii) Describe how that instrument is used in resource exploration. **2**
- (b) The diagram shows a map of the part of an ocean that includes two chains of features, a chain of islands and a chain of seamounts.



- (i) Name the geophysical phenomenon that accounts for the shape of the chain of islands. **1**
- (ii) Account for the formation and alignment of the chain of islands and the chain of seamounts. **3**
- (c) Describe how you carried out a first-hand investigation to determine the relationship between the nature of a surface and the radiation reflected from it. **4**
- (d) When the theory of plate tectonics was first proposed, some parts of the scientific community were reluctant to accept it. **6**
- Discuss the theory of plate tectonics and the evidence leading to its acceptance.
- (e) Discuss how information gathered from seismic observations has led to greater understanding of the structure of the Earth. **8**

Question 28 — Medical Physics (25 marks)

- (a) (i) Identify the purpose of a coherent bundle of optical fibres in an endoscope. **1**
- (ii) An optical fibre consists of a central core surrounded by cladding. Describe the role of the core and cladding. **2**

- (b) The table shows information relating to the transmission of sound through some types of body tissue.

<i>Tissue</i>	<i>Acoustic impedance</i> ($\times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$)	<i>Density</i> (kg m^{-3})	<i>Velocity of sound</i> (m s^{-1})
Muscle	1.70	1040	1630
Fat	1.38	945	1460
Bone	7.80	2560	3050

- (i) Identify ONE property of ultrasound. **1**
- (ii) Justify why, in an ultrasound scan, a boundary between muscle and bone would show up more clearly than would a boundary between muscle and fat. **3**
- (c) You have conducted a first-hand investigation to demonstrate the Doppler effect. Describe your investigation and conclusions. **4**
- (d) ‘CAT scans provide more information than X-rays, so they should be used whenever possible.’ Discuss this statement. **6**
- (e) Explain why MRI can be used to detect cancerous tissues. **8**

Question 29 — Astrophysics (25 marks)

- (a) (i) Define the term *binary stars*. **1**
- (ii) Describe the characteristics of its spectrum that identify a spectroscopic binary. **2**

- (b) The table shows information about three stars in the Milky Way galaxy.

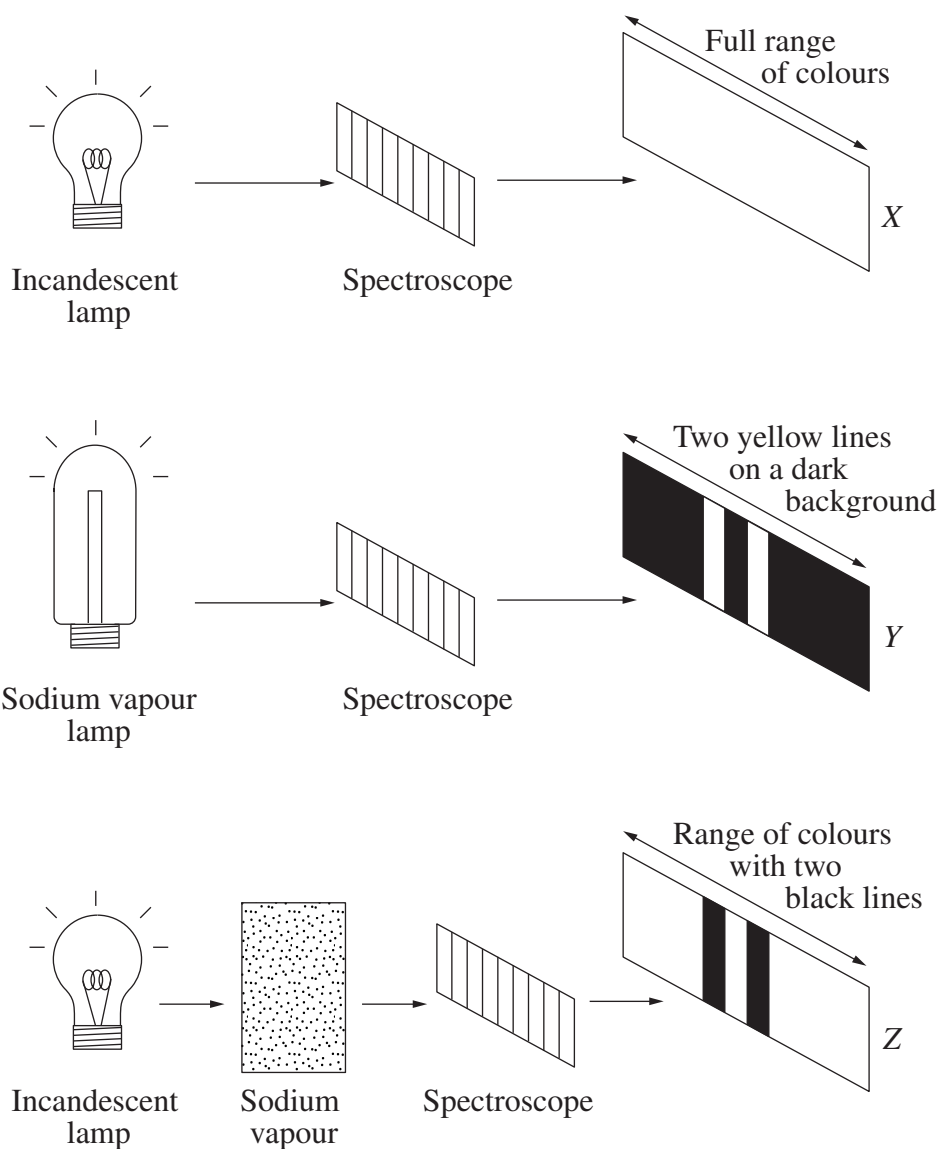
<i>Name</i>	<i>Spectral class</i>	<i>Distance from Sun (parsecs)</i>	<i>Apparent magnitude</i>
Betelgeuse	M2	184	+0.41
Achernar	B5	20	+0.47
Deneb	A2	429	+1.24

- (i) Identify which of the stars has the greatest surface temperature. **1**
- (ii) If Deneb and Betelgeuse were viewed from the same distance, which would appear brighter? Justify your answer. **3**

Question 29 continues on page 29

Question 29 (continued)

- (c) A student carried out an experiment to examine the spectra of various light sources through spectroscopes as shown in the diagram. The student observed three different spectra. 4



Account for the differences in the three observed spectra.

- (d) A new generation of Earth-based optical telescopes is advancing optical astronomy. Describe the advances in design that have been incorporated in large telescopes over recent years. 6
- (e) Explain how the data presented in Hertzsprung–Russell diagrams may be used to understand the evolution of stars. 8

End of Question 29

Question 30 — From Quanta to Quarks (25 marks)

- (a) (i) Define *nucleon*. 1
- (ii) Contrast ONE property of nucleons. 2

- (b) The table shows the quantum numbers of the four lowest states of the hydrogen atom, together with the energies of those states.

<i>Quantum number, n</i>	<i>Energy (joule)</i>
1 (Ground state)	0
2	1.63×10^{-18}
3	1.94×10^{-18}
4	2.04×10^{-18}

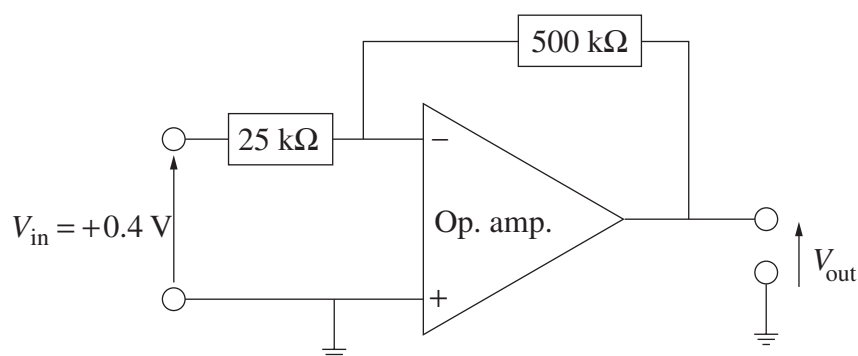
- (i) What is the energy of the photon emitted when an electron in the $n = 4$ level makes a transition to the $n = 3$ level? 1
- (ii) Use the data to draw the energy level diagram for hydrogen, and indicate on this diagram where the energy levels lie for quantum numbers greater than 4. 3
- (c) Describe how you carried out a first-hand investigation to determine the penetrating power of alpha, beta and gamma radiation on a range of materials. 4
- (d) The Manhattan Project is the codename given to the development of atomic (nuclear fission) bombs during World War II. 6

Discuss the significance of this project for society.

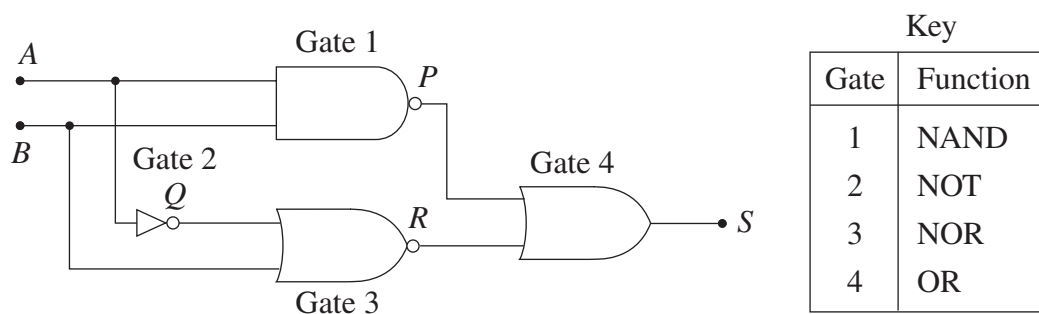
- (e) Analyse how Chadwick's and Fermi's work resulted in a greater understanding of the atom. 8

Question 31 — The Age of Silicon (25 marks)

- (a) (i) State the name of the transducer that is commonly used in a light meter of a camera. 1
- (ii) Describe the relationship between the amount of light incident on the transducer referred to in part (i), and its resistance. 2
- (b) An ideal differential-input operational amplifier is connected into the following circuit.



- (i) Explain the function of the 500 kΩ resistor in this circuit. 1
- (ii) Determine the output voltage, V_{out} . 3
- (c) A student constructed the following circuit in which four different logic gates were used. The circuit had two inputs, A and B , and one output, S . 4



For each of the possible input states of A and B , construct a truth table showing the output of Gate 1 at P , Gate 2 at Q , Gate 3 at R and Gate 4 at S .

- (d) Discuss the possibility that there may be a limit on the growth of computer power. 6
- (e) Discuss the impact that developments in electronics have had on society. 8

End of paper

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DATA SHEET

Charge on the electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2.0 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck's constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg's constant, R_H	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

FORMULAE SHEET

$$c = f\lambda$$

$$\text{Intensity} \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$F = - \frac{Gm_1m_2}{r^2}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$E = \frac{F}{q}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\frac{I_A}{I_B} = 100^{(m_B - m_A)/5}$$

$$d = \frac{1}{p}$$

$$v_{\text{av}} = \frac{\Delta s}{\Delta t}$$

$$a_{\text{av}} = \frac{\Delta v}{\Delta t} = \frac{v - u}{t}$$

$$\Sigma F = ma$$

$$E_k = \frac{1}{2}mv^2$$

$$p = mv$$

$$\Delta p = Ft$$

$$F = BIl \sin \theta$$

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$\tau = Fd$$

$$\tau = nBIA \cos \theta$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

FORMULAE SHEET

$$E_p = -\frac{Gm_1m_2}{r}$$

$$v = u + at$$

$$v_x^2 = u_x^2$$

$$v_y^2 = u_y^2 + 2a_y\Delta y$$

$$\Delta x = u_x t$$

$$\Delta y = u_y t + \frac{1}{2}a_y t^2$$

$$\frac{s}{t} = \frac{u+v}{2}$$

$$l_v = l_o \sqrt{1 - \frac{v^2}{c^2}}$$

$$t_v = \frac{t_o}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$F = qvB \sin \theta$$

$$E = \frac{V}{d}$$

$$E = hf$$

$$Z = \rho v$$

$$\frac{I_r}{I_o} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$\lambda = \frac{h}{mv}$$

$$\text{Amplifier gain} = \frac{V_{\text{out}}}{V_{\text{in}}}$$

$$A_o = \frac{V_o}{V_+ - V_-}$$

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																						
KEY																						
		Atomic Number		79 Au 197.0		Symbol of element																
		Atomic Weight		Gold		Name of element																
1 H 1.008 Hydrogen	4 Be 9.012 Beryllium	5 B 10.81 Boron	6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon															
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium	13 Al 26.98 Aluminium	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	18 Ar 39.95 Argon															
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.39 Zinc	31 Ga 69.72 Gallium	32 Ge 72.61 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton					
37 Rb 85.47 Rubidium	38 Sr 87.62 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.94 Molybdenum	43 Tc [98.91] Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon					
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57–71 Lanthanides		72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.8 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po [210.0] Polonium	85 At [210.0] Astatine	86 Rn [222.0] Radon				
87 Fr [223.0]	88 Ra [226.0]	89–103 Actinides		104 Rf [261.1]	105 Db [262.1]	106 Sg [263.1]	107 Bh [264.1]	108 Hs [265.1]	109 Mt [268]	110 Uun —	111 Uuu —	112 Uub —	113 Uuq —	114 Uuq —	115 —	116 Uuh —	117 —	118 Uuo —	Ununoctium			

Lanthanides

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [146.9] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinides

89 Ac [227.0] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237.0] Neptunium	94 Pu [239.1] Plutonium	95 Am [241.1] Americium	96 Cm [244.1] Curium	97 Bk [249.1] Berkelium	98 Cf [252.1] Californium	99 Es [252.1] Einsteinium	100 Fm [257.1] Fermium	101 Md [258.1] Mendelevium	102 No [259.1] Nobelium	103 Lr [262.1] Lawrencium
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Where the atomic weight is not known, the relative atomic mass of the most common radioactive isotope is shown in brackets.
The atomic weights of Np and Tc are given for the isotopes ²³⁷Np and ⁹⁹Tc.

2001 HSC Notes from
the Examination Centre
Physics

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2001 HSC NOTES FROM THE EXAMINATION CENTRE PHYSICS

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Physics. It provides comments with regard to responses to the 2001 Higher School Certificate Examination, indicating the quality of candidate responses, and highlighting the relative strengths and weaknesses of the candidature in each section and each question.

It is essential for this document to be read in conjunction with the relevant syllabus, the 2001 Higher School Certificate Examination, the Marking Guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Physics.

General comments

The 2001 HSC Physics paper is the first examination of the current syllabus, and was attempted by approximately 8945 candidates. It represents a shift in the style and nature of questions from those in previous Higher School Certificate papers, and the following general comments should be noted, regarding the manner in which these questions were handled by the candidature. Candidates did not always observe the instruction of the key verb within each question. This omission often resulted in a loss of marks, for example if a candidate only provided a description when an explanation was required.

In answering larger-mark holistic questions, candidates are expected to be able to communicate their responses in a coherent and logical manner. Many responses to these questions were unnecessarily long and inappropriate essays that did not address the question and often contained contradictory statements. Successful responses to these questions showed structure, often with some evidence of planning the response. A common feature of better responses was the use of diagrams and/or tables to present information. Point form presentation, when appropriate, neatly presents structured logic.

Teachers and candidates should be aware that examiners may ask questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course. This reflects the fact that the knowledge, understanding and skills, developed through the study of discrete sections, should accumulate to a more comprehensive understanding than may be described in each section separately.

Section I

Part A – Multiple choice

The following table shows the correct response to each question.

Question	Correct Response
1	D
2	A
3	A
4	C
5	C
6	B
7	B
8	D

Question	Correct Response
9	B
10	A
11	D
12	D
13	C
14	C
15	D

Part B

Specific Comments

Question 16

This question tested candidates' understanding of the time dilation effect of relativity, although some candidates did not recognise time dilation as the relevant concept. Most candidates were not able to manipulate the time dilation equation sufficiently well to achieve the correct answer.

Question 17

This question tested candidates' understanding of the vertical acceleration of rockets. The most common error made by candidates was in thinking that the graph showed uniform acceleration – some candidates stated this explicitly while a significant number of candidates attempted to evaluate the acceleration using equations of uniform acceleration. A large number of candidates correctly justified the increase of acceleration shown in the graph by describing the constant force and decreasing mass experienced by a rocket. A minority of candidates evaluated the gradient of the curve at the points indicated.

Question 18

This question tested candidates' understanding of projectile and circular motion. Part (a) involved a simple calculation and was handled well by most candidates. However, some candidates did not use the value of 'g' provided in the data sheet. The most common error in the calculation of part (b) was assuming that $u_y = 20 \text{ ms}^{-1}$, rather than zero. Another common error was not assigning similar directions to a_y and Δy . In the comparison required by part (c), most candidates did not recognise that gravitational force was common to both situations and that the centripetal force on B was the resultant of gravitational force and the tension force of the arm towards the pivot, and was larger than the force on A as a result.

Question 19

This question assessed candidates' understanding of the Michelson-Morley experiment and its connection with the theory of special relativity. Many candidates were able to discuss relativity in the terms required by the question, however the link between relativity and the Michelson-Morley experiment was generally not well expressed. Many candidates included points not relevant to the results of the Michelson-Morley experiment.

Question 20

This question tested candidates' understanding of transformers and electricity distribution. Many candidates were able to give correct reasons for stepping-up and stepping-down voltages in a distribution network. Generally, the use of step-down transformers was better understood than step-up transformers.

Question 21

This question tested candidates' understanding of back EMF in motors and its role in opposing the applied voltage to the motor. Many candidates failed to recognise the role of back EMF, instead giving detail of how a motor operates. Candidates who did recognise the role of back EMF were usually able to describe it clearly and relate this to the need for a resistance at start-up.

Question 22

This question tested candidates' understanding of parallel current-carrying wires in parts (a) and (b), and an investigation of the motor effect in part (c). Most candidates performed well in part (a), although some confused direction in relation to the diagram. In part (b), most candidates realised that the relationship to be graphed was linear, although few recognised the initial length of the wire as 3 metres.

In part (c), most candidates were able to provide a clear definition of the motor effect. Some candidates had difficulty distinguishing the motor effect from the motor. The investigations that demonstrated the motor effect were wide ranging, and many candidates could provide a correct result from an investigation, although most had difficulty using the result to explain the motor effect.

Question 23

This question tested candidates' knowledge of the effects electrical generators have had on society and the environment. The majority of candidates were able to identify relevant issues, although many did not identify or discuss a sufficient variety of both positive and negative issues. Most candidates' responses were unnecessarily lengthy and contained irrelevant information, such as re-stating the question and repetition of statements. The use of essay-style structure was not required to gain full marks. Responses made in point form or as a table were infrequent but usually scored well.

Question 24

This question tested candidates' understanding of the X-ray diffraction methods used by the Braggs. Many candidates chose vague and general areas eg medicine, without giving specific examples and purposes. Many candidates were unable to outline clearly the process of diffraction and the link between diffraction and determination of structure, though many candidates could link reflected and interfering rays to the structure of the crystal. Many candidates confused X-ray diffraction with electron diffraction, and the Braggs' experiments with those of other physicists. A common error was to make correct statements about interference without outlining how the Braggs used this to determine structure.

Question 25

This question tested candidates' understanding of experimental methods, based on the results of an experiment on the photoelectric effect. The drawing of the graph in part (a) was generally done well by the candidature. The most common errors were not placing the independent variable (frequency) on the horizontal axis, not drawing a line of best fit, and not using linear scales on the axes. In part (b), many candidates were able to explain that the reliability of the experiment could be improved by repetition and averaging.

Question 26

This question tested candidates' understanding of electrons and holes in semiconductors. The open-ended nature of this question meant that there was a wide range of valid responses. Candidates scored the full range of marks available, but only some candidates were able to score full marks on this question. A significant number of candidates were not able to explain the concept of a 'hole'. Some candidates confused semiconductors with superconductors.

Section II – Options

Specific Comments

Question 27 – Geophysics (attempted by approximately 1% of the candidature.)

- (a) This part examined knowledge of gravity survey instruments and was answered well by most candidates.
- (b) This part examined candidates' understanding of how a geological hot spot and a moving tectonic plate can produce an island chain. Most candidates did not identify the chain of islands as a hot spot path, instead interpreting the structure as an island arc and ignoring age.
- (c) This part examined candidates' application of experimental technique to geophysics, and was poorly handled by the candidature in general. General experimental techniques were poorly understood in this context. Some candidates focused their attention on nuclear, rather than electromagnetic radiation. Only some candidates were able to score full marks on the question.

- (d) This part required candidates to discuss the theory of plate tectonics and the evidence leading to its acceptance. Most candidates were able to score some marks on this question, by describing the theory of plate tectonics. Few candidates, however, discussed the evidence required by the question.
- (e) This part required candidates to discuss the link between seismic observations and increased knowledge of the structure of the Earth. Many candidates did not distinguish the meaning of 'seismic' observations. Demonstrated knowledge of the structure of the Earth, as well as seismic principles, was quite poor. Diagrams were especially helpful in answering this question, however the use of good diagrams was quite rare. Most diagrams presented were poorly drawn or not labelled.

Question 28 – Medical Physics (attempted by approximately 21% of the candidature.)

- (a) This part examined candidates' knowledge of optic fibres as used in endoscopes. Most candidates were able to identify the purpose of coherent bundles in the optic fibre. A common error in this part was to focus the response on incoherent bundles. Most candidates correctly identified that total internal reflection occurs at the core/cladding boundary, but did not describe the relative difference in refractive index.
- (b) This part examined candidates' understanding of the physical principles of ultrasound scans. Most candidates were able to identify one property of ultrasound, usually by stating that frequency $> 20\,000\text{ Hz}$. Some candidates were able to use the appropriate calculations to explain why some tissue boundaries show up more clearly than others. The most common error was to discuss the transmission of the ultrasound, not reflection. Another common error was failing to discuss how the ultrasound signal is detected.
- (c) This part examined candidates' application of experimental technique to medical physics, by focusing on the Doppler effect. A significant error made by many candidates was in not understanding the difference between loudness, frequency and wavelength. Many candidates said the loudness increased, rather than the pitch.
- (d) Most candidates correctly identified the nature of CAT scans. A number of candidates compared CAT scans and X-rays with other methods of diagnosis, rather than comparing CAT scans with X-rays.
- (e) Some candidates provided excellent answers to this question, however there was a variety of common errors. Very few candidates used the term 'resonance'. Many candidates failed to explain precession and what it is that actually precesses. Many candidates left out essential steps in the operation of MRI. A number of candidates said that the magnetic field was switched off.

Question 29 – Astrophysics (attempted by approximately 27.5% of the candidature.)

- (a) This was answered well by most candidates, though many had difficulty describing the concept of a common centre of mass. Many candidates correctly described red shifts and blue shifts, and the relative motions they represent. Few candidates, however, correctly described the periodic red shift/blue shift pattern characteristic of spectroscopic binaries.
- (b) Most candidates could identify the hottest star. The majority of better answers involved calculating the absolute magnitudes of the stars, and using this information to judge comparative luminosities. Candidates who tried to reason out an answer in words only, usually made errors or had insufficient steps of logic.
- (c) Many candidates demonstrated a basic understanding of the production of spectra, but only some candidates were able to express reasons for the difference between the spectra shown. Few candidates were able to describe the electron transitions that produce emission and absorption spectra. Some candidates simply restated information given in the stem of the question.
- (d) Many candidates provided answers that described developments outside the scope of the question. Many of those who did direct their response appropriately, demonstrated a basic recall of recent advances in telescope design. Only some candidates were able to describe correctly how these advances result in improved telescope performance.
- (e) Candidates had to relate each of the stellar evolutionary stages to the groups in the Hertzsprung-Russell diagram and known characteristics of those groups. Few candidates were able to do this well, often describing stellar evolution in detail without relating it correctly to the H-R diagram, or failing to describe the characteristics of each group. Some candidates gave detailed descriptions of stellar evolutionary stages that do not appear on the H-R diagram. Most candidates did not draw a H-R diagram and could only provide a poor description of it. Many candidates mistakenly believed that a ‘protostar’ is located at the right-hand top corner of the H-R diagram; and that stars move up or down the Main Sequence, from red to blue, or vice versa.

Question 30 – From Quanta To Quarks (attempted by approximately 48.5% of the candidature.)

- (a) The first part required candidates to define ‘nucleon’, and was generally not well answered. Many candidates confused nucleon with nucleus or neutron. Many candidates had difficulty contrasting protons and neutrons, that is, showing how they are different.
- (b) Most candidates were able to perform a subtraction to determine the energy of a photon produced by a particular transition. Many candidates presented alternative presentations of the information contained in the question. A common erroneous presentation was to draw electron orbits, often also adding the transitions of particular series, such as the Balmer and Paschen series.

- (c) Many candidates were able to describe an investigation to detect the penetrating power of different radiations including a discussion of the use of a control in this investigation. However, discussion of monitoring and improving the reliability of the investigation was generally poorly handled. Many candidates gave the impression they had done the investigation as a computer simulation. These candidates did not provide good discussions of the issues of reliability or the use of a control.
- (d) There was a wide variation in performance amongst the candidature. Most candidates provided responses that were insufficiently detailed. The more successful candidates on this question were conversant with the history of the period and gave a discussion appropriate to the scope of the question.
- (e) The majority of candidates were able to state that Chadwick discovered the neutron and many could describe the relevant experiment in detail and apply the law of conservation of momentum to indicate that the suspected gamma rays were in fact particles. Analyses of Fermi's work varied from quite detailed to limited, many believing that Fermi discovered the neutrino and nuclear fission. Many candidates made some attempt to answer this question, although there was often a lack of direction in the responses. A significant number of candidates answered the question as if Chadwick and Fermi worked together.

Question 31 – The Age Of Silicon (attempted by approximately 2% of the candidature.)

- (a) Most candidates were able to name the transducer commonly used in a light meter or a camera but were not generally able to identify whether the response of the transducer was linear or non-linear; and whether it was in direct or inverse proportionality to light intensity.
- (b) Many candidates were able to identify the function of the feedback resistor, but the calculation in the second part was generally poorly handled.
- (c) Generally, the candidature experienced a high level of success at deducing the output of a set of logic gates demonstrating a good understanding of the principles involved.
- (d) This question was answered quite poorly. Discussions included vague generalisations, such as stating that as technology improves, computers get faster, without examining the technical reasons for this improvement. Most candidates restricted themselves to discussing speed increases without looking at other elements of computer 'power'.
- (e) This part was generally poorly handled by the candidature. Many candidates confused the 'developments in electronics' with the 'use of electronics', and 'electronics' with 'electricity'. Hence, the weaker responses were simply a list of household appliances or electronic equipment.

Physics

2001 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I: Part A			
1	1	p.50; 9.2.1; Col. 2; DP 1: p.50; 9.2.1; Col. 3; DP 3	H6, H9
2	1	p. 55; 9.3.1; Col. 2; DP 1: p. 59; 9.4.1; Col. 2; DP 4	H9
3	1	p. 62; 9.4.4; Col. 3; DP 2	H3, H14
4	1	p. 56; 9.3.3; Col. 2; DPs 1, 3	H9
5	1	p. 51; 9.2.2; Col. 2; DPs 5, 8	H6, H9, H14
6	1	p. 60; 9.4.2; Col. 2; DPs 6, 7 p. 60; 9.4.2; Col. 3; DP 5	H8, H10
7	1	p. 51; 9.2.2; Col. 3; DP2	H6, H9
8	1	p. 56; 9.3.2; Col. 2; DP 7: p. 56; 9.3.2; Col. 3; DP 4	H4, H7, H9
9	1	p. 52; 9.2.3; Col. 2; DP 2; p. 52; 9.2.3; Col. 3; DP 1	H7, H9, H12
10	1	Prelim; p. 32; 8.3.2; Col.3; DP 5 p.56; 9.3.2; Col. 2; DPs 4, 5, 6	H9, H14
11	1	p. 57; 9.3.4; Col. 2; DPs 3, 4, 5	H4, H7, H9
12	1	p. 62; 9.4.4; Col. 2; DP 7	H10, H14
13	1	p. 51; 9.2.2; Col. 2; DP 8	H6
14	1	p. 55; 9.3.1; Col. 3; DP 3 p. 55; 9.3.1; Col. 2; DP 2	H9, H14
15	1	p. 51; 9.2.2; Col.2; DP 1	H6, H7, H9, H13, H14
Section I: Part B			
16(a)	1	p.53; 9.2.4; Col. 2; DP 11	H6
16(b)	3	p.53; 9.2.4; Col. 3; DP 5	H6
17(a)	2	Prelim; p. 38; 8.4.2; Col.2; DP 6 p. 51; 9.2.2; Col. 2; DP 8	H6, H14
17(b)	4	p. 51; 9.2.2; Col. 2; DP 8	H6, H14

Question	Marks	Content	Syllabus outcomes
18(a)	1	p.50; 9.2.1; Col.1 p.50; 9.2.1 Col.2, DP 1 p.50; 9.2.1 Col.3, DP3	H6, H9
18(b)	2	p.51; 9.2.2 Col3, DP2	H6, H9
18(c)	3	p.51; 9.2.2; Col.2, DPs 1, 2, 9 p.51; 9.2.2; Col.3, DPs 1, 2, 3 p.52; 9.2.2; Col.3, DP 1	H6, H9
19	4	p.53; 9.2.4; Col. 2; DPs 2, 3, 5, 6, 7, 11	H2, H6, H13, H14
20	4	p. 57; 9.3.4; Col. 1 p. 57; 9.3.4; Col. 2; DPs 1, 2, 4, 5 p. 57; 9.3.4; Col. 3; DP 4	H4, H7, H9, H14
21	3	p. 56; 9.3.2; Col. 2; DP 6	H7, H9, H13
22(a)	1	p. 55; 9.3.1; Col. 2; DP 3 p. 55; 9.3.1; Col. 3; DP 3	H9
22(b)	2	p. 55; 9.3.1; Col. 2; DPs 2, 3	H9, H13
22(c)	4	p. 55; 9.3.1; Col. 3; DP 2	H9
23	6	p. 56; 9.3.3; Col. 2; DP 5	H4, H13
24(a)	2	p. 62; 9.4.4; Col. 2; DP1	H1, H2, H3, H4
24(b)	4	p. 62; 9.4.4; Col. 2; DP1	H7, H8, H9
25(a)	4	p. 60; 9.4.2; Col. 2; DPs 6, 7	H13
25(b)	2	p. 46; 9.1 p. 47 p. 48	H11 H12 H14
26	8	p. 61; 9.4.3; Col. 2; DPs 2, 3, 4, 5; 8, 9 p. 61; 9.4.3; Col. 3; DPs.1, 2	H1, H8, H10, H13, H14

Question	Marks	Content	Syllabus outcomes
Section II			
Geophysics			
27(a)(i)	1	p. 65; 9.5.2; Col. 2, DPs 8, 11	H9
27(a)(ii)	2	p. 65; 9.5.2; Col. 2, DPs 8, 11	H9
27(b)(i)	1	p. 66; 9.5.4; Col. 2, DP 4	H1, H2, H14
27(b)(ii)	3	p. 66; 9.5.4; Col. 2, DP 4	H1, H2, H14
27(c)	4	p. 65; 9.5.2; Col. 3, DP 1	H8, H10, H12, H13
27(d)	6	p. 66; 9.5.4; Col. 2; DPs 4, 5	H1, H2, H4, H13, H14
27(e)	8	p. 66; 9.5.3; Col 1; p. 66; 9.5.5; Col. 3; DP 2	H1, H4, H8, H13, H14
Medical Physics			
28(a)(i)	1	p. 69; 9.6.2; Col. 2; DP 6	H8, H13
28(a)(ii)	2	p. 69; 9.6.2; Col. 2; DP 5	H8, H13
28(b)(i)	1	p. 68; 9.6.1; Col. 2; DP1	H1, H4
28(b)(ii)	3	p. 68; 9.6.1; Col. 2; DP 5 p. 68; 9.6.1; Col. 3; DP 5	H8, H14
28(c)	4	p. 68; 9.6.1; Col. 3; DP 6	H8, H12, H13
28(d)	6	p. 69; 9.6.2; Col. 2; DPs 3, 4 p. 69; 9.6.2; Col. 3; DP 2	H1, H4, H8, H10, H13, H14
28(e)	8	p. 70; 9.6.4; Col. 3; DPs1, 2	H1, H4, H9, H13, H14
Astrophysics			
29(a)(i)	1	p. 74; 9.7.5; Col. 2; DP 1	H9
29(a)(ii)	2	p. 74; 9.7.5; Col. 2; Dp 1	H7, H8, H9
29(b)(i)	1	p. 73; 9.7.3; Col2. DPs 3, 4, 5	H7
29(b)(ii)	3	p. 73; 9.7.4; Col. 2; DP 1 p. 73; 9.7.4; Col. 3; DP 1	H7, H8, H14
29(c)	4	p. 73; 9.7.3; Col. 2; DPs 1, 2, 3, 4 p. 73; 9.7.3; Col. 3; DPs 1, 3	H7, H8, H13, H14
29(d)	6	p. 72; 9.7.1; Col. 2; DP 5 p. 72; 9.7.1; Col. 3; DP 3, 4	H1, H4, H7, H13
29(e)	8	p. 74; 9.7.6; Col. 1 p. 74; 9.7.6; Col. 3; DPs2, 3	H1, H3, H7, H13

Question	Marks	Content	Syllabus outcomes
<i>From Quanta to Quarks</i>			
30(a)(i)	1	p. 77; 9.8.4; Col. 2; DP 2	H10
30(a)(ii)	2	p. 77; 9.8.4; Col. 2; DP 2	H10
30(b)(i)	1	p. 76; 9.8.1; Col 2; DP 2 p. 76; 9.8.1; Col 3; DP 2	H7, H8, H10
30(b)(ii)	3	p. 76; 9.8.1; Col 2; DP 2 p. 76; 9.8.1; Col 3; DP 2	H7, H8, H10
30(c)	4	p. 78; 9.8.5; Col. 3, DP 2	H11, H13
30(d)	6	p. 78; 9.8.5; Col. 3; DP 1	H1, H4, H13
30(e)	8	p. 77; 9.8.4; Col. 2; DP 1 p. 77; 9.8.4; Col. 3; DP 4	H1, H2, H10, H13, H14
<i>The Age of Silicon</i>			
31(a)(i)	1	p. 81; 9.9.3; Col. 2; DP 2	H7
31(a)(ii)	2	p. 81; 9.9.3; Col. 2; DP 2 p. 81; 9.9.3; Col. 3; DP 1	H7
31(b)(i)	1	p. 82; 9.9.6; Col. 2; DPs 3, 4	H2
31(b)(ii)	3	p. 82; 9.9.6; Col. 2; DPs 3, 4, 6, 7 p. 82; 9.9.6; Col. 3; DP 3	H4, H9, H14
31(c)	4	p. 82; 9.9.5; Col. 2; DP 1 p. 82; 9.9.5; Col. 3; DP 1, 2	H9, H13
31(d)	6	p. 82; 9.9.7; Col. 2; DPs 1, 2 p. 82; 9.9.7; Col. 3; DP 1	H3, H4, H9, H13, H14
31(e)	8	p. 80; 9.9.1; Col. 3; DP 1 p. 80; 9.9.1; Col. 2; DP 1, 2, 3	H1, H4, H9, H13

2001 HSC Physics Marking Guidelines

Question 16 (a) (1 mark)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Time dilation	1

Question 16 (b) (3 marks)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Calculates the correct answer (units necessary)	3
EITHER • Time dilation equation correctly stated. Then with t_v and t_0 incorrectly substituted (but with answer appropriate to incorrect substitution) and recognition that the answer is physically absurd ($\frac{v^2}{c^2} < 0$) OR • Algebraic errors (but with answer pertinent to incorrect manipulation)	2
EITHER • Time dilation equation (correct) OR • Incorrect use of equation leading to correct calculation ending with $\frac{v^2}{c^2} < 0$. No recognition of physical absurdity OR • Gives correct value but shows no working (units may be omitted)	1

Question 17 (a) (2 marks)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
- Acceleration can be determined from the slope of velocity versus time graph - The acceleration is greater 100 s after lift-off Note: <i>either</i> a visual determination of the slope <i>or</i> a measurement from the graph, is acceptable.	2
One only of the above points	1

Question 17 (b) (4 marks)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
Correctly explains the shape of both parts of the graph	4
EITHER - Correct explanation of the first part of the graph OR - Partly-correct explanation for the first part, second part of the graph correct	3
Provides a correct statement relating to the first part of the graph, second part correct	2
EITHER - A correct statement relating to the first part of the graph OR - A correct statement relating to the second part of the graph	1

Question 18 (a) (1 mark)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
Correctly calculates the force	1

Question 18 (b) (2 marks)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
• Uses the correct formula and substitutes correct values	2
• Identifies the correct formula	1

Question 18 (c) (3 marks)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
• Correctly describes the resultant forces on <i>A</i> and <i>B</i> and correctly deduces that the force on <i>B</i> is greater than the forces on <i>A</i>	3
EITHER • Correctly describes the resultant forces on <i>A</i> and <i>B</i> but has an incorrect comparison OR • Correctly describes the resultant force on either <i>A</i> or <i>B</i> (but not both) and makes a correct comparison	2
One of the following: • Correctly describes the resultant force on <i>A</i> OR • Correctly describes the resultant force on <i>B</i> OR • Gives a correct comparison	1

Question 19 (4 marks)

Outcomes assessed: H2, H6, H13, H14

MARKING GUIDELINES

Criteria	Marks
- An accurate understanding of the results of the Michelson-Morley experiment - An understanding of the Special Theory of Relativity - Demonstrates the link between the Michelson-Morley experiment and the Special Theory of Relativity	4
EITHER - Demonstrates some understanding of both the Michelson-Morley experiment and the Special Theory of Relativity OR - Demonstrates an accurate understanding of either the Michelson-Morley experiment or the Special Theory of Relativity	2–3
EITHER - Reference to the constant speed of light OR - Reference to no aether	1

Question 20 (4 marks)

Outcomes assessed: H4, H7, H9, H14

MARKING GUIDELINES

Criteria	Marks
- Clearly and correctly explains why transformers are used, using a range of reasons relating to: – function of transformers – energy loss in transmission – safety and environmental issues	4
- Clearly and correctly explains why transformers are used, referring to two of the three above areas OR - Demonstrates a basic knowledge of the three above areas	3
- Correctly explains why transformers are used, referring to one of the three above areas OR - Demonstrates a basic knowledge of two of the three above areas	2
States one reason	1

Question 21 (3 marks)

Outcomes assessed: H7, H9, H13

MARKING GUIDELINES

Criteria	Marks
A complete explanation involving the description of back emf in motors, outlining the need for protective resistance at low speed, and the production of back emf at higher speeds that opposes the current through the windings	3
An explanation, with reference to the back emf opposing the current through the windings. The back emf increases or decreases with the speed of the motor	2
Some reference to back emf	1

Question 22 (a) (1 mark)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Gives correct direction (attractive)	1

Question 22 (b) (2 marks)

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
EITHER - Graph showing linear relationship, starting at $1 = 3$ m indicated OR - Graph showing direct (linear) proportionality with $1 = 3$ m indicated	2
Graph showing a proportional relationship (linear relationship starting at origin)	1

Question 22 (c) (4 marks)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
The explanation of how the results demonstrate the motor effect displays a thorough knowledge of the physics of the motor effect and relates to the specific investigation they undertook	4
EITHER The explanation of how the results demonstrate the motor effect displays a sound knowledge of the physics of the motor effect and relates to the specific investigation they undertook	3
States the motor effect and describes their investigation. Provides no link between the two	2
EITHER - Describes their investigation OR - States the motor effect	1

Question 23 (6 marks)

Outcomes assessed: H4, H13

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough understanding of the effects of generators on society and the environment by discussing a positive and a negative aspect of at least one societal effect and at least one environmental effect	5–6
EITHER - Provides at least one positive aspect on both society and the environment indicating a thorough understanding of the issues OR - Provides at least one negative aspect on both society and the environment indicating a thorough understanding of the issues OR - Provides positive and negative aspects of at least one societal effect and at least one environmental effect, indicating a sound understanding of the issues	3–4
- States both one aspect of a societal issue AND an environmental issue OR - Indicates a sound understanding of a societal issue OR - Indicates a sound understanding of an environmental issue	2
States either one aspect of a societal issue or one aspect of an environmental issue	1

Question 24 (a) (2 marks)

Outcomes assessed: H1, H2, H3, H4

MARKING GUIDELINES

Criteria	Marks
BOTH - States an example in which an understanding of crystal structure is useful AND - A purpose for this use	2
EITHER - States an example in which an understanding of crystal structure is useful OR - A purpose for this use	1

Question 24 (b) (4 marks)

Outcomes assessed: H7, H8, H9

MARKING GUIDELINES

Criteria	Marks
Provides general features of the methods used by the Braggs and demonstrates a thorough knowledge of how crystal structure was determined	3–4
Provides general features of the methods used by Braggs and demonstrates a basic knowledge of how crystal structure was determined	1–2

Question 25 (a) (4 marks)

Outcomes assessed: H13

MARKING GUIDELINES

Criteria	Marks
- Provides labelled axes and indicates proper units - Provides a linear scale which is appropriately chosen - Correctly plots the data points - Draws a reasonable line of best fit which may, or may not, include the outlier point. (Neither dot-to-dot connecting nor curves are acceptable.)	4
One error or omission with the remaining features correct	3
Two errors or omissions with remaining features correct	2
One feature correct	1

Question 25 (b) (2 marks)

Outcomes assessed: H11, H12, H14

MARKING GUIDELINES

Criteria	Marks
• Provides examples of how to improve the reliability of the experiment	2
• Describes one way in which reliability can be improved	1

Question 26 (8 marks)

Outcomes assessed: H1, H8, H10, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates an extensive knowledge and understanding of the physics of electrons and holes through a well-connected explanation	7–8
• Demonstrates a knowledge of the major points relating to electrons and holes with an amplification of some of them. Communicates the ideas effectively	5–6
• Demonstrates a basic knowledge of some of the major points relating to electrons and holes. Communicates the ideas clearly	3–4
• States ONE or TWO of the points relating to electrons and holes	1–2

Question 27 (a) (i) (1 mark)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• Either <i>Gravity meter</i> or <i>Gravimeter</i>	1

Question 27 (a) (ii) (2 marks)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• A description is given of the information provided by the instrument and the scope of its use	2
EITHER • A description is given of the information provided by the instrument OR • A description is given of the use of the instrument	1

Question 27 (b) (i) (1 mark)

Outcomes assessed: H1, H2, H14

MARKING GUIDELINES

Criteria	Marks
One of: - Plate tectonics OR - Continental drift OR - Sea-floor spreading	1

Question 27 (b) (ii) (3 marks)

Outcomes assessed: H1, H2, H14

MARKING GUIDELINES

Criteria	Marks
Thoroughly explains both the formation and alignment with some correct reference to time	3
EITHER - Thoroughly explains both the formation and alignment but has no reference to time OR - Partly explains both the formation and alignment and has some correct reference to time	2
EITHER - Links existence of a hot-spot and plate motion OR - Relates the change in orientation of the island chain and seamount chain to change in motion of the plate	1

Question 27 (c) (4 marks)

Outcomes assessed: H8, H10, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Correctly describes an appropriate experiment with proper experimental procedures, including the specific points: - control of variables - repetition - measurement and observation	4
Correctly describes an appropriate experiment including ANY TWO of the specific points	3
Correctly describes an appropriate experiment including ONE of the specific points	2
Provides a partial description of an appropriate experiment	1

Question 27 (d) (6 marks)

Outcomes assessed: H1, H2, H4, H13, H14

MARKING GUIDELINES

Criteria	Marks
Discuss the issues, showing a detailed understanding of the theory and the evidence leading to its acceptance, including some discussion of the initial reservations	5–6
EITHER - Discusses the issues, showing a good understanding of the theory and some understanding of the evidence leading to its acceptance OR - Communicates some understanding of the theory and a good understanding of the evidence leading to its acceptance	3–4
States some aspects of either the theory <i>or</i> the evidence leading to its acceptance	1–2

Question 27 (e) (8 marks)

Outcomes assessed: H1, H4, H8, H13, H14

MARKING GUIDELINES

Criteria	Marks
Demonstrates extensive knowledge and understanding of how information gathered from seismic observations has led to greater understanding of the structure of the Earth in a coherent way	7–8
Demonstrates a thorough knowledge and understanding of how information gathered from seismic observations has led to greater understanding of the structure of the Earth	5–6
Demonstrates a basic knowledge and understanding of how information gathered from seismic observations has led to greater understanding of the structure of the Earth	3–4
Demonstrates a limited knowledge of the information gathered from seismic observations	1–2

Question 28 (a) (i) (1 mark)

Outcomes assessed: H8, H13

MARKING GUIDELINES

Criteria	Marks
Correct purpose identified	1

Question 28 (a) (ii) (2 marks)

Outcomes assessed: H8, H13

MARKING GUIDELINES

Criteria	Marks
Describes the role by EITHER - referring to both total internal reflection and that $n_{\text{core}} > n_{\text{cladding}}$ OR - drawing a well-labelled diagram	2
EITHER - refers to total internal reflection or $n_{\text{core}} > n_{\text{cladding}}$ OR - draws and unlabelled / poorly-labelled diagram	1

Question 28 (b) (i) (1 mark)

Outcomes assessed: H1, H4

MARKING GUIDELINES

Criteria	Marks
One property correctly identified	1

Question 28 (b) (ii) (3 marks)

Outcomes assessed: H8, H14

MARKING GUIDELINES

Criteria	Marks
EITHER - States that reflection of ultrasound occurs at a boundary between media with dissimilar acoustic impedances - The largest reflected energy results in the biggest signal returned to transducer. This occurs when the acoustic impedances are most dissimilar OR - Correct calculations of reflection coefficients with correct interpretations	3
EITHER - Gives a statement such as “reflection of ultrasound occurs most strongly at a boundary between media with the most dissimilar acoustic impedances” OR - Correct calculations of reflection coefficients without interpretation	2
EITHER - Gives a statement such as “reflection of ultrasound occurs at a boundary between media with dissimilar acoustic impedances” OR - Identifies correct equation and substitutes numbers but does not reach correct answer(s)	1

Question 28 (c) (4 marks)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
Complete description of investigation methods (results) and conclusion	3–4
Partial description of methods and /or conclusion	1–2

Question 28 (d) (6 marks)

Outcomes assessed: H1, H4, H8, H10, H13, H14

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of limitations and advantages of CAT scans as compared to traditional X-rays	5–6
Demonstrates a basic knowledge of limitations and advantages of CAT scans as compared to traditional X-rays	3–4
Mentions only one or two limitations or advantages of CAT scans or X-rays	1–2

Question 28 (e) (8 marks)

Outcomes assessed: H1, H4, H9, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Demonstrates an accurate understanding of the physical principles involved: magnetic fields, precession, resonance - Applies this understanding to explain why cancerous cells can be detected using MRI - Presents the information coherently	7–8
- Demonstrates some understanding of the physical principles involved in MRI - Relates this understanding to cancerous cells	5–6
Demonstrates some understanding of some of the physical principles involved in MRI	3–4
EITHER - Recalls the names of 1 or 2 of the physical principles involved: magnetic fields, precession, resonance OR - Identifies the water content of cancerous cells as their distinguishing feature	1–2

Question 29 (a) (i) (1 mark)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Correct answer	1

Question 29 (a) (ii) (2 marks)

Outcomes assessed: H7, H8, H9

MARKING GUIDELINES

Criteria	Marks
• Describes the periodic red/blue shift of the spectrum	2
• Mentions the red/blue shift in the spectrum	1

Question 29 (b) (i) (1 mark)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
• Correct answer	1

Question 29 (b) (ii) (3 marks)

Outcomes assessed: H7, H8, H14

MARKING GUIDELINES

Criteria	Marks
• Correctly calculates M for both stars • Correctly relates absolute magnitude to brightness • Explains that Deneb's more negative number means it is brighter	3
• Correctly calculates M for ONE star • Correctly relates absolute magnitude to brightness	2
EITHER • Incorrectly calculates M for both stars, and • Correctly relates absolute magnitude to brightness OR • Statement that Deneb has more negative absolute magnitude than Betelgeuse	1

Question 29 (c) (4 marks)

Outcomes assessed: H7, H8, H13, H14

MARKING GUIDELINES

Criteria	Marks
Identifies the differences and provides correct reasons for those differences	3–4
Indicates a basic knowledge and understanding of the reasons for the differences	1–2

Question 29 (d) (6 marks)

Outcomes assessed: H1, H4, H7, H13

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of recent advances in telescope design. The answer should indicate how these developments led to improvements in the resolution, sensitivity, or cost-effectiveness of telescopes	5–6
Demonstrates a basic knowledge of recent advances in telescope design. The answer should give some indication as to how these developments led to improvements in the resolution, sensitivity, or cost-effectiveness of telescopes	3–4
Mentions only one or two recent advances in telescope design	1–2

Question 29 (e) (8 marks)

Outcomes assessed: H1, H3, H7, H13

MARKING GUIDELINES

Criteria	Marks
In a coherent discussion, demonstrates an extensive knowledge of how the H–R diagram may be used to understand the evolution of stars	7–8
Demonstrates a thorough knowledge of how the H–R diagram may be used to understand the evolution of stars	5–6
Demonstrates a sound knowledge of how the H–R diagram may be used to understand the evolution of stars	3–4
Demonstrates a basic knowledge of how the H–R diagram may be used to understand the evolution of stars	1–2

Question 30 (a) (i) (1 mark)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
EITHER • Both nuclear particles named OR • Statement such as “Nucleons are the particles in the nucleus”	1

Question 30 (a) (ii) (2 marks)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Correctly identifies ONE property and correctly states the difference	2
• Identifies ONE property that differs. (In the case of charge, the charge on ONE particle must be correctly indicated eg. positive or neutral)	1

Question 30 (b) (i) (1 mark)

Outcomes assessed: H7, H8, H10

MARKING GUIDELINES

Criteria	Marks
• Gives correct answer	1

Question 30 (b) (ii) (3 marks)

Outcomes assessed: H7, H8, H10

MARKING GUIDELINES

Criteria	Marks
EITHER - Sketch of energy level diagram with line spacings (approx.) correct; quantum numbers ($n = 1-4$) and energies of levels identified with some indication of where energy levels lie for $n > 4$ OR - Sketch of energy level diagram, with line spacings (approx.) correct, series limit or location of levels for $n > 4$ identified, but with EITHER quantum numbers OR energies of levels omitted	3
EITHER - Sketch of energy level diagram, with line spacings (approx.) correct, but with EITHER quantum numbers OR energies of levels omitted OR - Sketch of energy level diagram, with line spacings even or similarly inappropriate, but quantum numbers and energies of levels identified	2
EITHER - Sketch of energy level diagram, with line spacings even or similarly inappropriate, and EITHER quantum numbers OR energies of levels omitted OR - Sketch of energy level diagram, with line spacings (approx.) correct, but with NEITHER quantum numbers NOR energies of levels identified	1

Question 30 (c) (4 marks)

Outcomes assessed: H11, H13

MARKING GUIDELINES

Criteria	Marks
- Correctly describes an appropriate experiment with proper experimental procedures, including the specific points: - control of variables - repetition - measurement and observation	4
Correctly describes an appropriate experiment including ANY TWO of the specific points	3
Correctly describes an appropriate experiment including ONE of the specific points	2
Provides a partial description of an appropriate experiment	1

Question 30 (d) (6 marks)

Outcomes assessed: H1, H4, H13

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of the positive and negative effects of the Manhattan Project on society	5–6
Demonstrates a sound knowledge of the positive and negative effects of the Manhattan project on society	3–4
Mentions only one or two effects of the Manhattan Project on society	1–2

Question 30 (e) (8 marks)

Outcomes assessed: H1, H2, H10, H13, H14

MARKING GUIDELINES

Criteria	Marks
Answer demonstrates an extensive knowledge of the work of both scientists, expressed in a coherent way, and showing an analysis of how their work contributed to an understanding of the atom	7–8
Answer includes a thorough description of the work of both scientists, and how their work led to an understanding of the atom	5–6
EITHER - Answer shows a basic knowledge of the work of both scientists OR - Answer demonstrates a sound knowledge of the work of one scientist	3–4
EITHER - Answer shows a limited knowledge of the work of both scientists OR - Answer shows a basic knowledge of the work of one scientist	1–2

Question 31 (a) (i) (1 mark)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
Correct answer	1

Question 31 (a) (ii) (2 marks)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
Mathematical relationship or a statement of that relationship or a graph showing the relationship	2
Gives a statement such as “Resistance goes down with increasing light level”	1

Question 31 (b) (i) (1 mark)

Outcomes assessed: H2

MARKING GUIDELINES

Criteria	Marks
Provides an explanation of the function of the 500 kΩ resistor	1

Question 31 (b) (ii) (3 marks)

Outcomes assessed: H4, H9, H14

MARKING GUIDELINES

Criteria	Marks
Correct output voltage (sign must be correct)	3
Uses a correct calculation method, but gets wrong answer	2
EITHER - Statement about virtual earth OR - At least ONE step in derivation	1

Question 31 (c) (4 marks)

Outcomes assessed: H9, H13

MARKING GUIDELINES

Criteria	Marks
Table all correct	4
Output from 3 of the 4 gates correct, consistent with whatever error has been made	3
Output from 2 gates correct, consistent with other errors	2
Output from 1 gate correct, consistent with other errors	1

Question 31 (d) (6 marks)

Outcomes assessed: H3, H4, H9, H13, H14

MARKING GUIDELINES

Criteria	Marks
Demonstrates a thorough knowledge of limitations to the growth of computer power and how they arise. Mentions alternative(s) to current technologies.	5–6
Demonstrates a basic knowledge of limitations to the growth of computer power and reasons (may mention alternative technologies)	3–4
EITHER - Mentions only a few limitations, OR - Discusses computing power in terms of either speed or memory only	1–2

Question 31 (e) (8 marks)

Outcomes assessed: H1, H4, H9, H13

MARKING GUIDELINES

Criteria	Marks
Demonstrates an extensive knowledge and describes a number of key developments in electronics and discusses, discussing their social impacts in a coherent way	7–8
Demonstrates a thorough knowledge and describes some of the key developments in electronics and discusses their social impacts	5–6
Demonstrates a sound understanding of the way in which electronics has affected society, but mentions only a few examples of developments in electronics or does not have a coherent discussion	3–4
Provides only a couple of examples of developments in electronics or provides a cursory discussion of their social impact or does not make clear the impacts on society	1–2